Optimizers in Quantum Approximate Optimization Algorithm (QAOA) for the Vehicle Routing Problem

Overview and Motivation

The Quantum Approximate Optimization Algorithm (QAOA) is a hybrid quantum-classical algorithm that finds approximate solutions to complex combinatorial optimization problems. A major application for QAOA is the **Vehicle Routing Problem (VRP)**, a logistics challenge that seeks the most efficient routes for a fleet of vehicles. The VRP is computationally difficult, with complexity that grows exponentially with the problem size.

QAOA addresses this by translating the VRP into a quantum operator called a **cost Hamiltonian**. The lowest energy state of this Hamiltonian corresponds to the optimal VRP solution. The process is executed in a feedback loop:

- A quantum computer prepares and measures a special quantum state.
- A classical computer uses these measurement results to adjust the quantum circuit's parameters to improve the solution.

This write-up focuses on the crucial role of the **optimizer** in the classical part of this loop. The optimizer's purpose is to find the optimal values for the QAOA's two sets of parameters: $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$. These angles, which control the application of the mixer and cost Hamiltonians, are essential for the quality of the final solution.

Optimizer Explanation

Optimizers are broadly categorized into two types: classical and quantum.

- **Classical Optimizers:** These are algorithms that run on a classical computer to control the quantum part of the QAOA.
- **Quantum Optimizers:** A newer class of optimizers designed specifically to leverage quantum information to better navigate the unique landscape of quantum state spaces. An example is the **Quantum Natural Gradient (QNG)**.

Classical optimizers can be further divided into two groups: **gradient-based** and **gradient-free** methods.

- **Gradient-Based Optimizers:** These methods use the gradient (the direction of the steepest descent) to guide their search for the minimum. Think of it as a hiker always taking a step in the steepest downward direction to get to the bottom of a valley.
- **Gradient-Free Optimizers:** These methods do not use gradient information. Instead, they probe the surrounding landscape by making small changes to parameters and checking if the solution improves. This is like a person in a foggy valley, trying a few steps in different directions and choosing the one that leads down.

Examples of well-known optimizers include:

- **Gradient-Based:** SPSA (Simultaneous Perturbation Stochastic Approximation), Adam, and BFGS (Broyden–Fletcher–Goldfarb–Shanno algorithm).
- **Gradient-Free:** COBYLA (Constrained Optimization by Linear Approximation) and Nelder-Mead.

Optimizer Choosing

Choosing an optimizer for QAOA on VRP is a practical decision that depends on the problem size and available hardware. A classical optimizer is typically used today because quantum optimizers are still an area of active research.

For QAOA applied to VRP, a **gradient-free** method is often a more useful choice.

- **Why a Gradient-Free Optimizer?**
  The QAOA parameter landscape is highly non-convex and noisy with many local minima. Attempting to compute a reliable gradient on a Noisy Intermediate-Scale Quantum (NISQ) device is challenging. Gradient-free methods are more robust to this noise and are less likely to get stuck in barren plateaus, where gradients vanish.
- **Why COBYLA?**
  COBYLA is a good choice for smaller VRP instances because it is designed to handle non-linear optimization with constraints, which is how the VRP is encoded in the QAOA circuit. It is a derivative-free method, making it well-suited for noisy environments.
- **Drawbacks of COBYLA:**
  The main drawback of COBYLA is that it can be **slow to converge**. As the number of parameters (determined by the QAOA circuit depth) increases, the search space becomes much larger. Without the guidance of a gradient, finding the optimal solution can become very time-consuming.